Molecular communication between Intermuscular Adipose Tissue and Skeletal Muscle under progressing Insulin Resistance.

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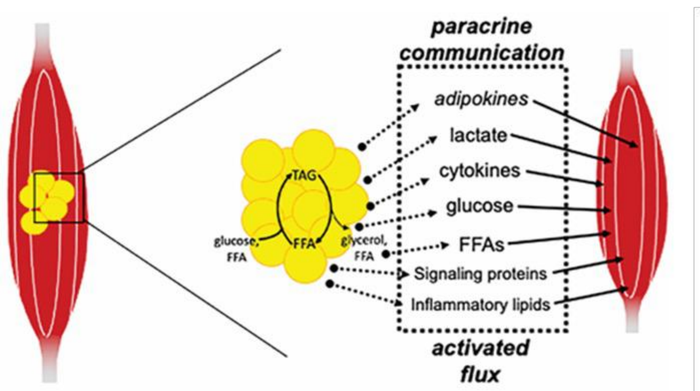
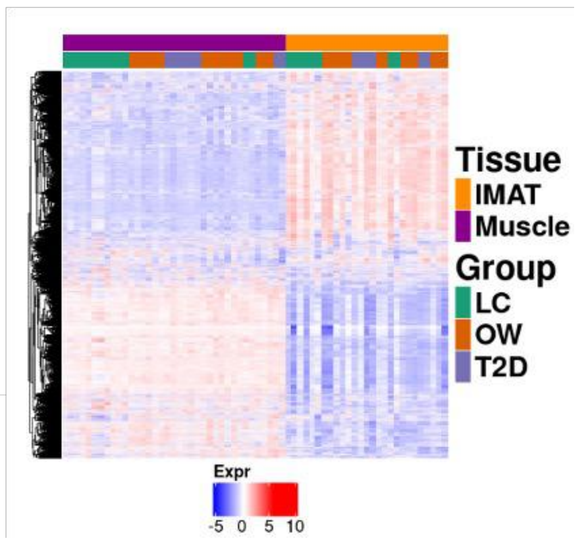
Hypothesis: Differentially expressed(DE) Intermuscular Adipose Tissue (IMAT) secretomes and their receptors in the Skeletal Muscle(SM) modulate SM insulin sensitivity.

IMAT may regulate skeletal muscle IS

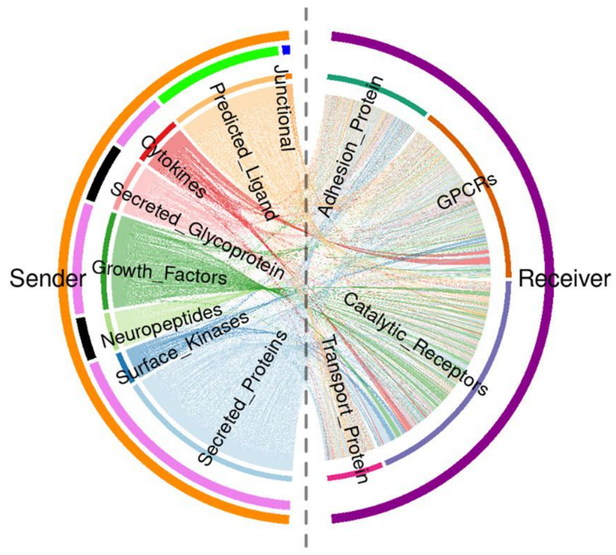


Sparks et al,. 2021, Diabetes

mRNA expression level



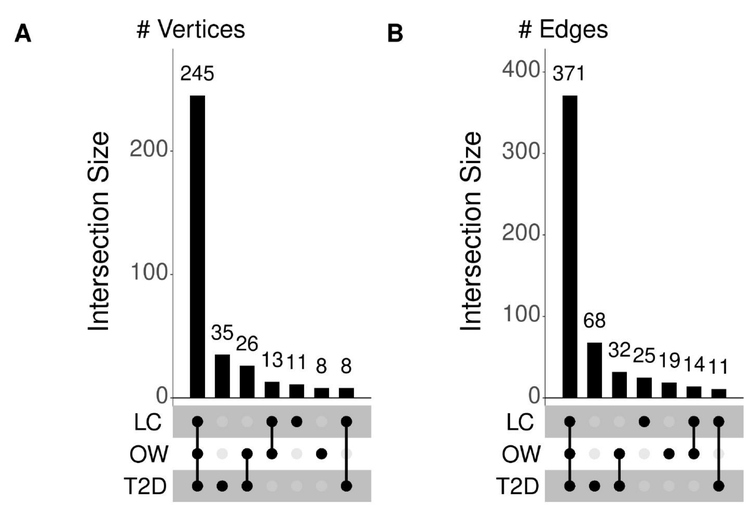
Sender receiver database(SRD)

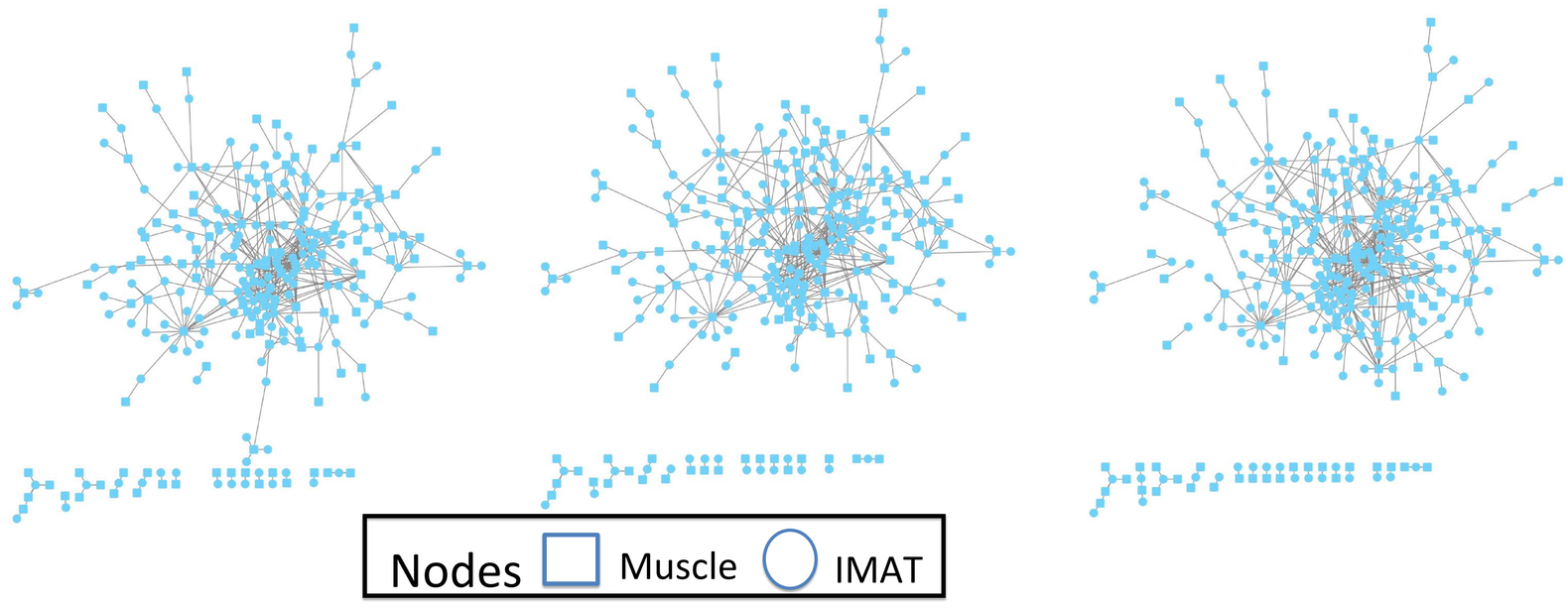


- Over 15 database
- 1194 Sender(S)
- 1402 Receiver(R)
- 9877 SR pairs
- 1541 Endocrine
- 6277 Paracrine
- 117 Juxtacrine
- 1942 Predicated ligands

Higher number nodes and edges is observed in the T2D network.

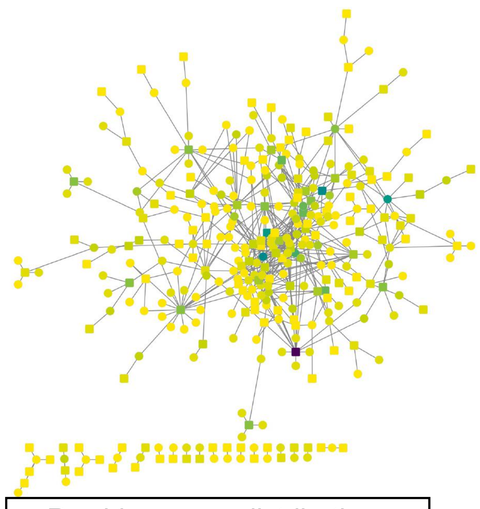
The number of nodes(A) and edges (B)



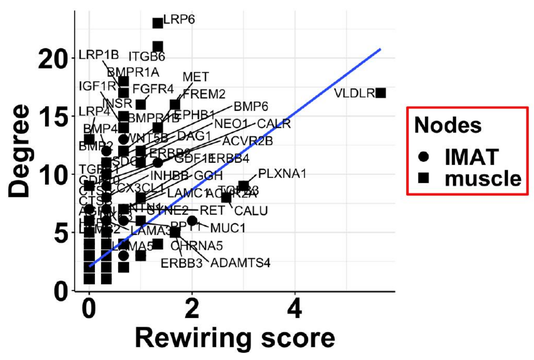


The rewired and hub genes were muscle specific and functionally correlated to metabolic disorders.

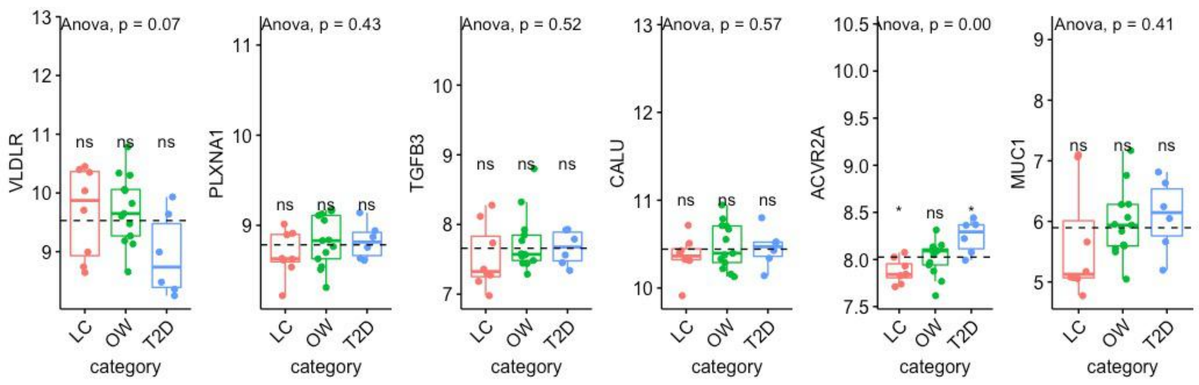
Rewiring analysis using Dynet(Ivan HG et al,. 2016, Bioinformatics)



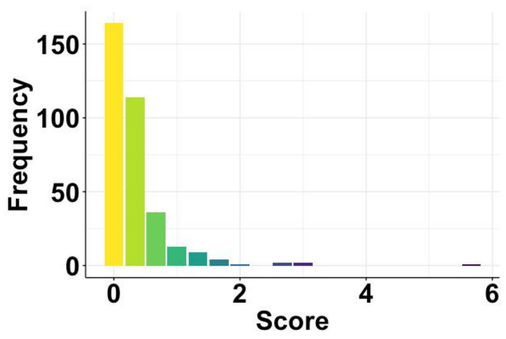
Correlations between degree and rewiring



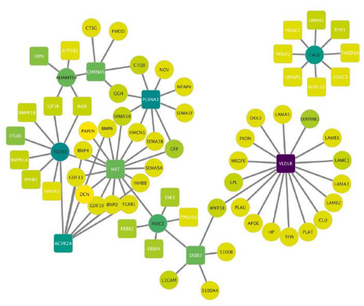
Expression level of the top six most rewired genes in IMAT



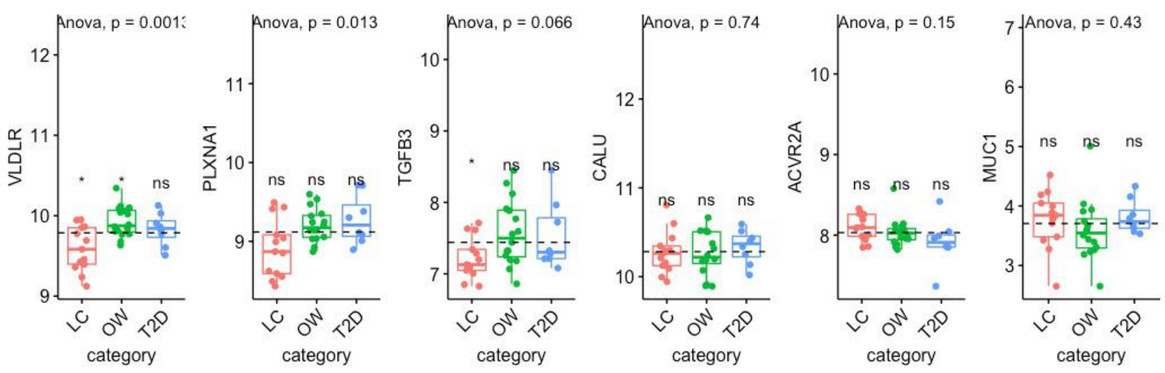
Rewiring score distributions



Top ten most rewired genes



Expression level of the top six most rewired genes in the skeletal muscle



Summary

- Communication from IMAT to muscle is augmented as insulin resistance advance.
- The rewired and hub genes were muscle specific and functionally correlated to metabolic disorders.
- T2D specific IMAT genes are enriched to functions such as lipoprotein assembly, remodelling, and clearance, and disease associated to glycosylation, and lipid storage.